

PR-Centric Voice Manager + Autonomous Agent Workflow

Operating Manual (OpenHands + OpenRouter)

Updated target: build Agent Mesh Observatory & Governor as a Shadow AI Control Plane

Last updated: Dec 25, 2025

Key idea: iOS Shortcuts is only a voice UI. All real work happens in GitHub (state) and the agent runner (compute). This workflow keeps agents productive while preserving tight safety and review gates.

1. Components and responsibilities

Phone + iOS Shortcuts (voice UI)

- Capture dictation (AirPods / phone mic).
- Call GitHub REST API to create PRs, post PR comments, and set PR labels.
- Never executes code; never runs the agent.

GitHub (control plane)

- PR is single source of truth for each feature.
- Labels implement a simple state machine.
- Comments are the command channel (you <-> agents).
- CI checks are the hard quality gate before merge.

GitHub Actions (compute)

- Runs OpenHands in isolated runner, checks out PR branch, executes steps, runs tests.
- Commits + pushes to PR branch only; posts run summary back to PR.
- Never merges automatically.

OpenHands + OpenRouter (agent + models)

- OpenHands orchestrates role agents (backend, frontend, SDK, docs, tests).
- OpenRouter routes to a default model + fallback list to control cost/reliability.

2. PR-centric contract

- **One PR per feature** (small, reviewable increments; target 1-3 hours of agent work).
- **PR template required:** goal, definition of done, constraints, current status, next steps, and questions (yes/no or MCQ only).
- **State machine via labels:** agent-takeover (run), needs-input (pause + ask), paused (stop), ready-to-merge (human final gate).

Recommended labels

- agent-takeover, needs-input, paused, ready-to-merge
- security-review (auth/secrets/policy engine changes), perf, infra

3. Default safety rules

- Agents can only push to PR branches; **never** merge to main.
- No production credentials in agent runs. Use mocked/stubbed integrations until V0 is stable.
- Directory fences: constrain agents to the module(s) touched by the PR.
- CI must be green + you approve before merge.
- If the same error repeats twice: set needs-input and ask MCQ options.

Tool access policy for the repo

- Allowed: git (commit/push), package manager install, unit tests/lint, local docker compose.
- Disallowed by default: deploy keys, cloud credentials, payment actions, destructive DB operations.

4. Build plan as PRs (starter pilot)

Validate the workflow with 3-5 small PRs:

- Repo scaffold + CI + devcontainer / docker compose.
- Minimal ingest API + canonical event schema + Postgres tables (events + decision logs).
- Minimal UI page: recent events + basic filters (env/team/tool).
- Policy service MVP: JSON policy rules + allow/deny + rationale codes.
- PEP wrapper MVP: a tiny SDK/proxy that calls PDP before executing a dummy tool.

Success criteria for pilot

- You can create PRs and answer questions entirely by voice.
- Agent completes at least one PR end-to-end without manual coding.
- Costs and runtime are bounded (no runaway loops).

5. OpenRouter model policy (cost + reliability)

- Pick ONE default model for consistent behavior.
- Have 1-2 fallbacks for hard tasks or repeated failures.
- Log tokens used per agent run in the PR summary comment.
- Set a monthly budget cap; fail closed when exceeded.

6. Voice console: minimum shortcuts

- **New Feature PR:** dictate feature name + goal + DoD + constraints; create draft PR + label agent-takeover.
- **Reply to PR (voice):** dictate PR number + answer; post comment; optionally remove needs-input.
- **Set PR State:** dictate pause/resume/ready-to-merge; apply/remove labels.

7. Day-to-day cadence

- Morning (2-5 min): choose 1-3 PRs to progress; label agent-takeover.
- During the day: answer needs-input questions by voice; resume with agent-takeover.
- Evening (10-20 min): review ready-to-merge PRs; merge; queue next PRs.

8. How this maps to the product

- We are building the **control plane** first (policy + identity + decision logs).
- Then the **data plane** (PEPs: user proxy + agent SDK/proxy).
- Then the **IR plane** (timeline + evidence bundle export).